MTH 202 Chapter 5: Topics In Vector Calculus

5.5 Surface Integrals

Motivation:
In section 5.2.1 we discussed the idea that a line integral was useful for adding up a quantity over a given curve *C*. In this section we wish to extend this idea to adding up a quantity over a given *surface* σ to create the idea of a surface integral.

This idea can be useful in a variety of applications

- Adding up charge across a magnetized surface.
- Adding up gravitational forces across a surface.
- Adding up the density across a surface to help find the mass of the surface.
- Adding up flow of fluid through σ to help find the total quantity of fluid flowing through σ.

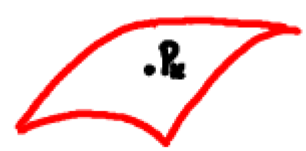
So, how do we construct an integral that will do this?

Exploration: Suppose we have a smooth surface σ that we plan to move across, and we want to add up the quantity at each point given by the function $f(x,y,z)$. Let Q represent the total quantity over the surface. How do we find Q?

We can begin by partition σ up into n patches.

Consider the k^{th} patch and choose a representative point $P_k\left(x_k^*, y_k^*, z_k^*\right)$ for this patch.

If we let ΔS_k represent the surface area of this patch and Q_k represent the quantity that exists over this patch. If we measure function *f* at $P_k\left(x_k^*, y_k^*, z_k^*\right)$ we obtain an approximation of the quantity we are adding up over this patch to be $Q_k \approx f\left(x_k^*, y_k^*, z_k^*\right)\Delta S_k$



Therefore, $Q \approx \sum_{k=1}^{n} f\left(x_k^*, y_k^*, z_k^*\right)\Delta S_k$ and as n approaches infinity we suspect the approximation to approach the correct value.

Therefore, we have $Q = \lim_{n\to\infty}\sum_{k=1}^{n} f\left(x_k^*, y_k^*, z_k^*\right)\Delta S_k = \iint_{\sigma} f(x,y,z)\,dS$

This leads us to the following definition.

Definition: Assuming that f is continuous, the <u>surface integral of f over the surface σ</u> is $\iint_{\sigma} f(x,y,z)\,dS$.

Remark: Let $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$ be the parameterization of the surface σ. We saw in section 4.4 that the surface area of the k^{th} patch, ΔS_k, is approximated by $\Delta S_k \approx \|\mathbf{r}_u \times \mathbf{r}_v\|\,dA$.

Therefore, we can rewrite the previous definition as follows.

Definition (Updated): Assuming that f is continuous, the <u>surface integral of f over the surface σ</u> is

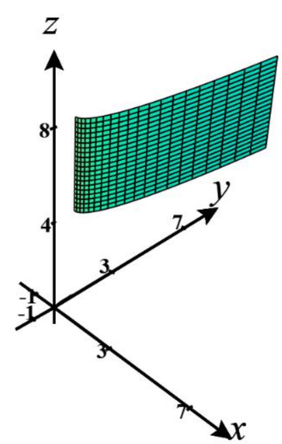
$$\iint_{\sigma} f(x,y,z)\,dS = \iint_{D} f(\mathbf{r}(u,v))\|\mathbf{r}_u \times \mathbf{r}_v\|\,dA.$$

Remark: Note that if $f(x,y,z) = 1$, then the surface integral $\iint_{\sigma} f(x,y,z)\,dS$ becomes $\iint_{\sigma} dS = \iint_{D} \|\mathbf{r}_u \times \mathbf{r}_v\|\,dA$ which we know from section 4.4 yields the surface area of σ.

<u>**Example 1:**</u> Consider the surface $\sigma = \{(x,y,z) \mid y = 2x^2 + 1,\ 0 \le x \le 2,\ 4 \le z \le 8\}$.

a) Find a parameterization $\mathbf{r}(u,v)$ for σ.

A parameterization for σ is $\mathbf{r}(u,v) = \langle u, 2u^2+1, v\rangle$ where $0 \le u \le 2,\ 4 \le v \le 8$.



b) Evaluate $\iint_{\sigma} (xz^2)\,dS$.

We note that $\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 4u & 0 \\ 0 & 0 & 1 \end{vmatrix} = \langle 4u, -1, 0\rangle$. Then, $\|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{16u^2+1}$, and we have that

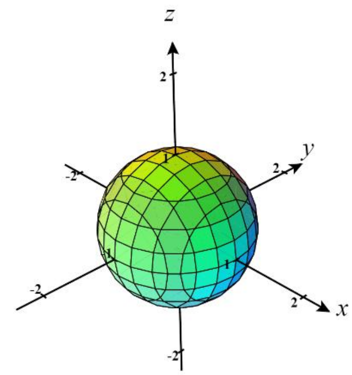
$$\iint_{\sigma} (xz^2)\,dS = \int_4^8 \int_0^2 uv^2\sqrt{16u^2+1}\,du\,dv = \int_4^8 \left[\left(\frac{1}{48}(16u^2+1)^{\frac{3}{2}}\right)\Bigg|_{u=0}^{u=2}\right] v^2\,dv$$

$$= \int_4^8 \frac{65\sqrt{65}-1}{48} v^2\,dv = \frac{65\sqrt{65}-1}{144} v^3 \Bigg|_4^8$$

$$= \frac{28}{9}\left(65\sqrt{65}-1\right).$$

Example 2: Compute the surface integral $\iint_{\sigma} z^2 dS$ where σ is the unit sphere $x^2+y^2+z^2=1$.

Hint: Use what you know about spherical coordinates to find parametric equations for σ.



Parametric equations for σ are:

$x=\sin(\phi)\cos(\theta)$ $y=\sin(\phi)\sin(\theta)$ $z=\cos(\phi)$ where $0\le\phi\le\pi,\ 0\le\theta\le2\pi$.

If we replace ϕ with u and θ with v we can parameterize the surface as

$\mathbf{r}(u,v)=\langle\sin(u)\cos(v),\sin(u)\sin(v),\cos(u)\rangle$ where $0\le u\le\pi,\ 0\le v\le2\pi$.

Note that you don't have to replace ϕ *with u and* θ *with v and then get*

$\mathbf{r}(\phi,\theta)=\langle\sin(\phi)\cos(\theta),\sin(\phi)\sin(\theta),\cos(\phi)\rangle$ where $0\le\phi\le\pi,\ 0\le\theta\le2\pi$.

In order to set up the surface integral as an iterated integral we need to know $\|\mathbf{r}_u\times\mathbf{r}_v\|$. So, we calculate that

$$\mathbf{r}_u\times\mathbf{r}_v=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \cos(u)\cos(v) & \cos(u)\sin(v) & -\sin(u)\\ -\sin(u)\sin(v) & \sin(u)\cos(v) & 0\end{vmatrix}$$

$$=\langle\sin^2(u)\cos(v),\sin^2(u)\sin(v),\cos(u)\sin(u)\cos^2(v)+\cos(u)\sin(u)\sin^2(v)\rangle$$

$$=\langle\sin^2(u)\cos(v),\sin^2(u)\sin(v),\cos(u)\sin(u)\rangle.$$

Then,

$$\|\mathbf{r}_u\times\mathbf{r}_v\|=\sqrt{\sin^4(u)\cos^2(v)+\sin^4(u)\sin^2(v)+\sin^2(u)\cos^2(u)}$$

$$=\sqrt{\sin^4(u)+\sin^2(u)\cos^2(u)}=\sqrt{\sin^2(u)\left(\sin^2(u)+\cos^2(u)\right)}=|\sin(u)|.$$

Since we know that $0\le u\le\pi$, we have that $|\sin(u)|=\sin(u)$.

Now, we have that $\iint_{\sigma} z^2\,dS=\int_0^{2\pi}\int_0^{\pi}(\cos(v))^2\sin(u)\,du\,dv=\int_0^{2\pi}\int_0^{\pi}\sin(u)\cos^2(u)\,du\,dv$. If we let $k=\cos(u)$, then we note that $dk=-\sin(u)du\Rightarrow-dk=\sin(u)du$, $k(0)=1$, and $k(\pi)=-1$. This means

$$\int_0^{2\pi}\int_0^{\pi}\sin(u)\cos^2(u)\,du\,dv$$

$$=\int_0^{2\pi}\int_1^{-1}-k^2\,dk\,dv=\int_0^{2\pi}\left[\left(-\frac{k^3}{3}\right)\Bigg|_{k=1}^{k=-1}\right]dv$$

$$=\int_0^{2\pi}\left(\frac{1}{3}+\frac{1}{3}\right)dv=\int_0^{2\pi}\frac{2}{3}\,dv=\frac{2}{3}(2\pi)=\frac{4\pi}{3}.$$

Example 3: Suppose σ is a smooth surface given by $z = g(x,y)$ where $(x,y) \in D$. Express the surface integral $\iint_{\sigma} f(x,y,z)\,dS$ as a double integral in terms of *x* and *y*.

Using a technique from section 4.4, we know that a parameterization for σ is: $\mathbf{r}(u,v) = \langle u, v, g(u,v)\rangle$ where *u* and *v* satisfy the same conditions as *x* and *y*. We can therefore rewrite this parameterization as: $\mathbf{r}(x,y) = \langle x, y, g(x,y)\rangle$. By definition we know that $\iint_{\sigma} f(x,y,z)\,dS = \iint_{D} f(x,y,g(x,y))\left\|\mathbf{r}_x \times \mathbf{r}_y\right\| dA$.

We now have written the surface integral as a double integral in terms of *x* and *y*, but we will simplify the double integral a bit further. Note that $\mathbf{r}_x = \langle 1, 0, g_x(x,y)\rangle$ and $\mathbf{r}_y = \langle 0, 1, g_y(x,y)\rangle$. This implies that

$$\mathbf{r}_x \times \mathbf{r}_y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & g_x(x,y) \\ 0 & 1 & g_y(x,y) \end{vmatrix} = \langle -g_x(x,y), -g_y(x,y), 1\rangle \text{ and } \left\|\mathbf{r}_x \times \mathbf{r}_y\right\| = \sqrt{\left(g_x(x,y)\right)^2 + \left(g_y(x,y)\right)^2 + 1}.$$

So, $\iint_{D} f(x,y,g(x,y))\left\|\mathbf{r}_x \times \mathbf{r}_y\right\| dA = \iint_{D} \left[f(x,y,g(x,y))\sqrt{\left(g_x(x,y)\right)^2 + \left(g_y(x,y)\right)^2 + 1} \right] dA.$

Note: In this example we are assuming that f is continuous on σ. Oftentimes this condition is just assumed without being stated in the textbook.

The most recent example is the justification for the following formulas.

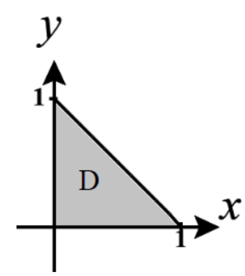
Formulas: Suppose σ is a smooth surface given by $z = g(x,y)$ where $(x,y) \in D$. Then,

1. A normal vector to the surface at $(x, y, g(x,y))$ is $\langle -g_x(x,y), -g_y(x,y), 1\rangle$.
2. $\iint_{\sigma} f(x,y,z)\,dS = \iint_{D} \left[f(x,y,g(x,y))\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \right] dA$

Note: It is possible to derive similar formulas in the cases where the surface is given by $y = h(x,z)$ or when the surface is given by $x = j(y,z)$. Can you guess what the formulas would be in these cases? See page 1133 of the book for one of the formulas.

Example 4: Fully setup, but do not evaluate, the surface integral $\iint_{\sigma} 2\cos(y)(z-\tan(y)+1)\,dS$ where σ is the portion of the surface $z=\tan(y)$ which exists over the triangular region with endpoints $(0,0)$, $(1,0)$, & $(0,1)$.

We first note that the region D has the following picture and can be defined as $D=\{(x,y)|0\le x\le 1 \ \&\ 0\le y\le 1-x\}$ know that



$$\iint_{\sigma} 2\cos(y)(z-\tan(y)+1)\,dS = \iint_{D} 2\cos(y)(\tan(y)-\tan(y)+1)\sqrt{(0)^2+\sec^2(y)+1}\,dA$$
$$= \int_0^1\int_0^{1-x} 2\cos(y)\sqrt{\sec^2(y)+1}\,dy\,dx$$

We now state a fact which we will recognize as an extension of a property of line integrals.

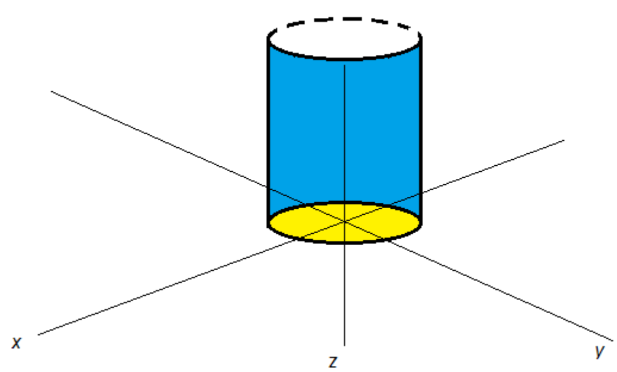
Fact: If σ is a finite union of smooth surfaces: $\sigma_1, \sigma_2, \ldots, \sigma_n$ (we say σ is <u>piecewise-smooth</u>), then

$$\iint_{\sigma} f(x,y,z)\,dS = \iint_{\sigma_1} f(x,y,z)\,dS + \iint_{\sigma_2} f(x,y,z)\,dS + \cdots + \iint_{\sigma_n} f(x,y,z)\,dS.$$

Example 5: Suppose that σ is the "can without a top" which consists of $\sigma_1=\{(x,y,z)\,|\,x^2+y^2=1, 0\le z\le 2\}$ and $\sigma_2=\{(x,y,z)\,|\,x^2+y^2\le 1,\ z=0\}$.

a) Find a parameterization $\mathbf{r}(u,v)$ of σ_1.

First, a picture of the surface with σ_1 in blue and σ_2 in yellow is:



Using the idea of cylindrical coordinates an answer is: $\mathbf{r}(u,v)=\langle \cos(u), \sin(u), v\rangle$ where $0\le u\le 2\pi$ and $0\le v\le 2$.

b) Find a parameterization $\mathbf{s}(u,v)$ of σ_2.

Using the idea of polar coordinates, an answer is: $\mathbf{s}(u,v)=\langle v\cos(u), v\sin(u), 0\rangle$ where $0\le u\le 2\pi$ and $0\le v\le 1$.

c) Evaluate $\iint_{\sigma}\left(x^2+y^2+z^2\right)dS$.

Using the most recent fact, we have that:

$$\iint_{\sigma}\left(x^2+y^2+z^2\right)dS=\iint_{\sigma_1}\left(x^2+y^2+z^2\right)dS+\iint_{\sigma_2}\left(x^2+y^2+z^2\right)dS$$

$$=\int_0^{2\pi}\int_0^2\left(\cos^2(u)+\sin^2(u)+v^2\right)\left\|\mathbf{r}_u\times\mathbf{r}_v\right\|dvdu+\int_0^{2\pi}\int_0^1\left(v^2\cos^2(u)+v^2\sin^2(u)\right)\left\|\mathbf{s}_u\times\mathbf{s}_v\right\|dvdu$$

$$=\int_0^{2\pi}\int_0^2\left(1+v^2\right)\left\|\mathbf{r}_u\times\mathbf{r}_v\right\|dvdu+\int_0^{2\pi}\int_0^1\left(v^2\right)\left\|\mathbf{s}_u\times\mathbf{s}_v\right\|dvdu.$$

So, we calculate: $\left\|\mathbf{r}_u\times\mathbf{r}_v\right\|=\left\|\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-\sin(u)&\cos(u)&0\\0&0&1\end{vmatrix}\right\|=\left\|\langle\cos(u),\sin(u),0\rangle\right\|=1$ and

$$\left\|\mathbf{s}_u\times\mathbf{s}_v\right\|=\left\|\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-v\sin(u)&v\cos(u)&0\\\cos(u)&\sin(u)&0\end{vmatrix}\right\|=\left\|\langle0,0,-v\rangle\right\|=v.$$

We are now able to calculate the answer:

$$\int_0^{2\pi}\int_0^2\left(1+v^2\right)\left\|\mathbf{r}_u\times\mathbf{r}_v\right\|dvdu+\int_0^{2\pi}\int_0^1\left(v^2\right)\left\|\mathbf{s}_u\times\mathbf{s}_v\right\|dvdu=\int_0^{2\pi}\int_0^2\left(1+v^2\right)dvdu+\int_0^{2\pi}\int_0^1\left(v^3\right)dvdu$$

$$=\int_0^{2\pi}\left(v+\frac{v^3}{3}\right)\Bigg|_{v=0}^{v=2}du+\int_0^{2\pi}\left(\frac{v^4}{4}\right)\Bigg|_{v=0}^{v=1}du=\int_0^{2\pi}\frac{14}{3}du+\int_0^{2\pi}\frac{1}{4}du=2\pi\left(\frac{14}{3}+\frac{1}{4}\right)=2\pi\left(\frac{59}{12}\right)=\frac{59\pi}{6}.$$

d) Note that σ_2 didn't *need* to be parameterized and could have just been expressed as the surface in the form of $z=f(x,y)$ over a region *D*. Find the surface integral for σ_2 using this approach.

We see that if σ_2 is written as $z=0$ over a region *D*, the we define $D=\{(r,\theta)|0\le r\le1\ \&\ 0\le\theta\le2\pi\}$.

Therefore,

$$\iint_{\sigma_2}\left(x^2+y^2+z^2\right)dS=\iint_D\left(x^2+y^2+(0)^2\right)\sqrt{(0)^2+(0)^2+1}\,dA=\int_0^{2\pi}\int_0^1 r^2\sqrt{1}\,rdrd\theta$$

$$=\int_0^{2\pi}\int_0^1 r^3drd\theta=\int_0^{2\pi}\frac{1}{4}d\theta=\frac{\pi}{2}$$